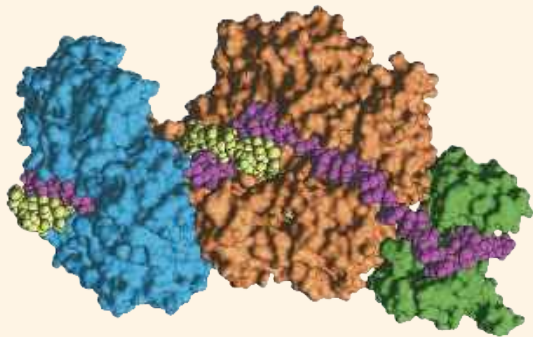


Levels 3-4: Applying/Analyzing

6. *E. coli* cells grown on ^{15}N medium are transferred to ^{14}N medium and allowed to grow for two more generations (two rounds of DNA replication). DNA extracted from these cells is centrifuged. What density distribution of DNA would you expect in this experiment?
- (A) one high-density and one low-density band
(B) one intermediate-density band
(C) one high-density and one intermediate-density band
(D) one low-density and one intermediate-density band
7. A student isolates, purifies, and combines in a test tube a variety of molecules needed for DNA replication. When she adds some DNA to the mixture, replication occurs, but each DNA molecule consists of a normal strand paired with numerous segments of DNA a few hundred nucleotides long. What has she probably left out of the mixture?
- (A) DNA polymerase (C) Okazaki fragments
(B) DNA ligase (D) primase
8. The spontaneous loss of amino groups from adenine in DNA results in hypoxanthine, an uncommon base, opposite thymine. What combination of proteins could repair such damage?
- (A) nuclease, DNA polymerase, DNA ligase
(B) telomerase, primase, DNA polymerase
(C) telomerase, helicase, single-strand binding protein
(D) DNA ligase, replication fork proteins, adenyl cyclase
9. **MAKE CONNECTIONS** Although the proteins that cause the *E. coli* chromosome to coil are not histones, what property would you expect them to share with histones, given their ability to bind to DNA (see Figure 5.14)?

Levels 5-6: Evaluating/Creating

10. **EVOLUTION CONNECTION** Some bacteria may be able to respond to environmental stress by increasing the rate at which mutations occur during cell division. How might this be accomplished? Might there be an evolutionary advantage to this ability? Explain.
11. **SCIENTIFIC INQUIRY**

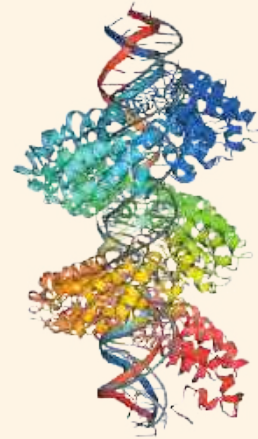


DRAW IT Model building can be an important part of the scientific process. The illustration shown above is a computer-generated model of a DNA replication complex. The parental and newly synthesized DNA strands are color-coded differently, as are each of the following three proteins: DNA pol III, the sliding clamp, and single-strand binding protein.

- (a) Using what you've learned in this chapter to clarify this model, label each DNA strand and each protein.
(b) Draw an arrow to indicate the overall direction of DNA replication.

12. **WRITE ABOUT A THEME: ORGANIZATION** The continuity of life is based on heritable information in the form of DNA, and structure and function are correlated at all levels of biological organization. In a short essay (100–150 words), describe how the structure of DNA is correlated with its role as the molecular basis of inheritance.

13. **SYNTHESIZE YOUR KNOWLEDGE**



This image shows DNA (gray) interacting with a computer-generated model of a TAL protein (multicolored), one of a family of proteins found only in a species of the bacterium *Xanthomonas*. The bacterium uses proteins like this one to find specific gene sequences in cells of the organisms it infects, such as tomatoes, rice, and citrus fruits. Given what you know about DNA structure and considering the image above, discuss how the TAL protein's structure suggests that it functions.

For selected answers, see Appendix A.